

RUBRICS FOR COURSE BASED DESIGN PROJECT

BCSE332L- Deep Learning

Phase I (Review -1) in moodle – (10 Marks)

•Focus: Problem Identification The core of this report is to identify a relevant real-world problem within the selected domain and articulate clear objectives for addressing it. Each student must define two distinct objectives that reflect their approach to the problem.

•Requirements: Domain Selection: Clearly specify the domain chosen (e.g., healthcare, education, agriculture, finance, etc.). Objective Framing: For the identified problem, each student should formulate two actionable objectives. Justification: Demonstrate how the problem is relevant to real-world scenarios. Justify the need for these objectives based on an analysis of the problem and its implications.

•Report submission in Moodle: Introduction (1 Page): Briefly introduce the problem, the domain, and its relevance to society or industry. Study Section (1 Page): Include a comparative analysis table with the following components:

- Describe the strengths and limitations of existing models.
- Highlight any novel observations or data trends.
- Mention the deep learning techniques applied (if any) in the domain.

Objectives & Justification: Present two objectives for each student and explain how they address the problem effectively. Link the justification to the pros and cons of existing models to show the need for new insights or improvements and provide standard data set.

Phase II (Review -2) in moodle – (10 Marks)

- Focus: DL model design, hyper parameter tuning
- Requirements: Design diagram, different models deployed, result comparison based on hyper parameter tuning.
- Presentation: video demonstration can be done in stream and upload the link in moodle.

Final Submission in vtop – (10 Marks)

- Focus: comparative analysis with different model and finding error metrics. Justification states how your work is best.
- Requirement Model Comparison: Analyze and compare multiple models applied to your problem domain.
- Error Metrics Evaluation: Include performance metrics (e.g., accuracy, F1 score) to show how well each model performs.
- Justification: Clearly explain why your solution stands out. Support it using the metric results and advantages of your chosen method.

- Final Report as per the format comprises of (Report submission in vtop): combine phase I and phase II along with all analysis, comparison tables, metrics, and justification. Final report should have 10 pages without plagiarism and individual student citation.

Note: Delay submission in moodle /vtop attract penalty.

Phase	Teaching Practice	Evaluation Method	Marks
Phase I	Project-Based Learning, Research-Based Learning, Flipped Classroom	Literature Review, Comparative Analysis, Research Gap Identification	10
Phase II	AI Pair Programming, Collaborative Coding, Design Thinking, Hands-on Lab	Model Design, Hyperparameter Tuning, Video Demonstration	10
Final Phase	Comparative Research, Scientific Writing, Reflective Learning	Model Comparison, Error Metrics, Technical Report, Viva	10
Total			30

Phase I: Problem Identification & Literature Review (Review-I) – Moodle (10 Marks)

Criteria	Marks
Problem Identification	2
Literature Review & Comparative Analysis	2
Research Gap Identification	2
Objectives & Justification	2
Dataset Selection & Report Quality	2
Total	10

Phase II: Model Design & Experimental Study (Review-II) – Moodle (10 Marks)

Criteria	Marks
Model Architecture Design	2
Implementation of Multiple Models	2
Hyperparameter Tuning	2
Experimental Analysis	2
Video Presentation & AI Pair Programming Reflection	2
Total	10

Final Submission – VTOP (Review 3) (10 Marks)

Criteria	Marks
Comparative Analysis	2
Error Metrics & Performance Evaluation	2
Justification of Proposed Model	2
Technical Report Quality & Citations	2
Individual Viva, Reflection & Demonstration	2
Total	10
